PLANT DISEASE DETECTION USING IMAGE PROCESSING AND DEEP LEARNING

Project Report — Image Processing and Computer Vision (IPCV)

Submitted by:
Team Member 1: Vartika(20233306)
Team Member 2: Vaishnavi(20233301)
Team Member 3: Yashshree(20233324)
Team Member 4: Yamini(20233320)

Department: Computer Science and Engineering
Institution: Motilal Nehru National Institute of Technology,Allahabad

1. Abstract

Plant diseases can reduce crop quality and productivity, making early detection important for preventing agricultural losses. This project presents a plant disease detection system that combines image processing and deep learning to analyse plant leaf images. CIELAB colour conversion and K-Means clustering support image segmentation, while morphological operations help remove unwanted noise. A MobileNetV2-based model classifies leaf images into healthy and diseased categories, and Grad-CAM visualizes the regions influencing predictions. A Streamlit interface allows users to upload leaf images and view results. The system covers nine classes of tomato, potato, and bell pepper leaves, demonstrating the application of computer vision in agriculture.

2. Introduction

Plant diseases can significantly lower crop yield and quality. Visual identification of these infections manually is often labor-intensive and demands specialized knowledge. To address this, the proposed system leverages computer vision and deep neural networks to examine leaf samples, integrating CIELAB colour conversion, K-Means clustering, MobileNetV2-based classification, Grad-CAM feature localization, and an interactive Streamlit web application for demonstrating classification outcomes.

2.1 Problem Statement

Manual identification of plant diseases can be time-consuming and may require expert knowledge. The objective is to develop a computer vision system that can analyse plant leaf images, classify them into supported healthy or diseased categories, and display the prediction in an understandable format.

2.2 Objectives

The main objectives of this project are:

- To apply image processing techniques to plant leaf images.
- To use K-Means clustering for grouping similar image pixels.
- To reduce small unwanted noise using morphological operations.
- To classify leaf images using a MobileNetV2-based deep learning model.
- To visualize important prediction regions using Grad-CAM.
- To provide a simple interface for testing the system with leaf images.

3. Methodology

The project combines image processing, deep learning, and visualization to detect plant diseases from leaf images.

3.1 Dataset Collection

The project uses a subset of the PlantVillage dataset containing approximately 12,335 images across nine classes of healthy and diseased tomato, potato, and bell pepper leaves. The dataset is divided into training (70%), validation (15%), and testing (15%) sets, as reported in the repository.

3.2 Image Preprocessing

Images are converted from RGB to CIELAB colour space, which separates lightness from colour information and helps analyse leaf colours for segmentation.

3.3 K-Means Clustering

K-Means groups similar pixels into three clusters (K = 3), helping separate visually different regions such as the background and leaf areas.

3.4 Morphological Opening

Morphological opening combines erosion and dilation to remove small unwanted noise from the segmented image and improve its clarity.

3.5 Disease Classification Using MobileNetV2

MobileNetV2 is a lightweight convolutional neural network that learns image features and classifies leaf images into the supported healthy and diseased categories.

3.6 Grad-CAM Visualization

Grad-CAM generates a heatmap highlighting image regions that influence the model's prediction, helping users understand where the model focused.

3.7 Application Interface

A Streamlit interface allows users to upload leaf images and view the predicted disease category and other available visualizations.

4. Results and Discussion

The project demonstrates an integrated workflow for plant leaf image analysis, disease classification, and prediction visualization. It combines traditional image processing techniques with a deep learning model and an interactive application interface.

4.1 Reported Performance

According to the project's repository, the model achieved the following results:

Evaluation parameter	Reported value
Dataset size	Approximately 12,335 images
Number of classes	9
Training split	70%
Validation split	15%
Testing split	15%
Test images	1,851
Test accuracy	99.78%
Macro F1-score	0.998

Accuracy represents the proportion of test predictions that are correct. The macro F1-score calculates the F1-score for each class and averages the scores equally, helping evaluate performance across different categories.

A confusion matrix can also be used to examine the classification results. It compares actual classes with predicted classes and helps identify which disease categories may be confused with one another.

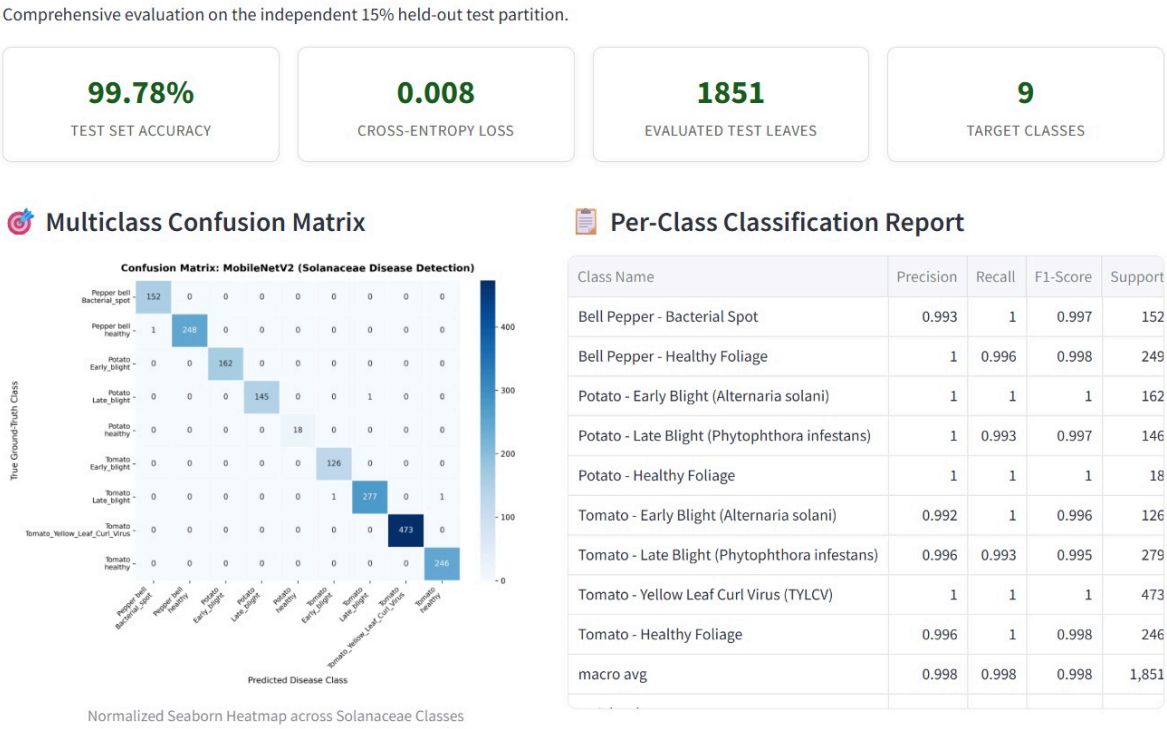


Figure 2. Model Evaluation Results and Multiclass Confusion Matrix

4.2 Discussion of Image Processing

CIELAB conversion and K-Means clustering help separate visually different leaf regions, while morphological opening removes small unwanted noise. However, segmentation accuracy may vary with lighting and background conditions.

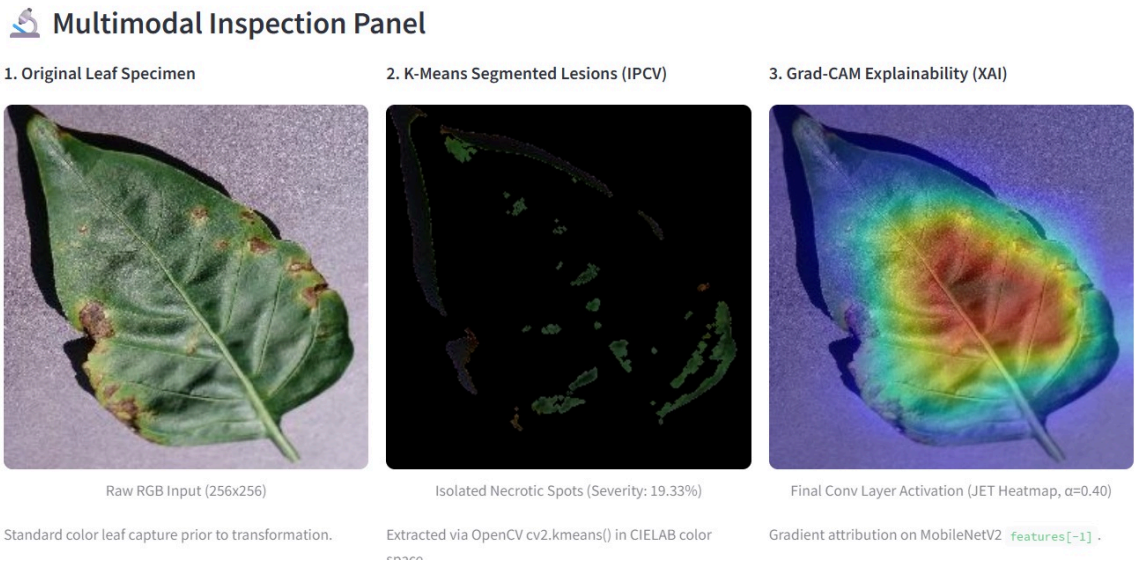


Figure 1: shows the original leaf image, the segmented regions obtained using K-Means clustering, and the Grad-CAM heatmap highlighting the regions influencing the model's prediction.

4.3 Discussion of Classification and Explainability

MobileNetV2 efficiently classifies leaf images, while Grad-CAM highlights the regions influencing predictions. However, real-world accuracy may vary due to lighting, background conditions, and image quality. Since the system supports only nine classes, its predictions are preliminary and should not replace expert diagnosis.

4.4 Limitations and Future Scope

The main limitations include the restricted number of disease classes, possible segmentation errors, variations in real-world image conditions, and the possibility of incorrect model predictions.

Future improvements could include training and evaluating the model on a larger and more diverse dataset, testing it on field photographs, adding more crops and disease categories, and improving segmentation under different lighting conditions. The application could also be extended to provide carefully verified information about disease prevention and treatment.

5. Conclusion

The Plant Disease Detection project demonstrates how digital image processing and deep learning can be combined to analyse plant leaf images. CIELAB colour-space conversion and K-Means clustering support image segmentation, while morphological opening helps reduce small unwanted regions. MobileNetV2

performs the classification of supported healthy and diseased leaf categories, and Grad-CAM visualizes the image regions that influence the model's prediction.

The Streamlit interface brings these components together in an accessible application. The project provides practical exposure to important IPCV concepts, including colour-space transformation, clustering, segmentation, morphological operations, deep learning-based classification, and visual interpretation of model predictions.

Although the system shows promising reported results, further testing on diverse real-world images is necessary before relying on it for practical agricultural decisions. Overall, the project illustrates the potential of computer vision as a supportive tool for preliminary plant disease identification.